



Laundry Pickup System

CS 385: Software Engineering

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1. Chapter 1

1.1 System descriptions:

In today's fast-paced world, where convenience is highly valued, managing laundry services efficiently has become essential. The **Laundry Pickup System** makes it easier for customers to handle their laundry needs without leaving the comfort of their homes. The system offers a variety of features, such as creating and managing accounts, scheduling pickups and deliveries, selecting service types, and making secure online payments. Customers can also track the status of their orders in real-time and provide ratings or feedback once the service is completed.

The system enhances the overall laundry experience by ensuring smooth communication between customers, laundry providers, and delivery drivers. Laundry providers can efficiently manage orders, update statuses, handle pricing and services, assign deliveries, and generate detailed reports. Integration with external delivery providers ensures timely pickups, accurate delivery updates, and a seamless service flow.

1.2 The development Process model used:

The **Agile Development Model** is the most effective approach for developing the Laundry Pickup System. This method supports iterative and incremental development, allowing the team to deliver the system in manageable stages. Each phase focuses on implementing specific features, including account management, order creation, tracking, payment processing, delivery coordination, and reporting.

Using this model, the development team can gather feedback from users after each iteration and make necessary adjustments before proceeding to the next stage. This process ensures that the system continuously adapts to user needs and expectations, resulting in a **reliable, efficient, and user-friendly platform** for laundry service management.

1.3 Functional Requirements:

User Requirements

1. Customer shall be able to create an account.
2. Customer shall be able to log in.
3. Customer shall be able to edit their profile information.
4. Customer shall be able to specify location.
5. Customer shall be able to create a new laundry order.
6. Customer shall be able to track their order status.
7. Customer shall be able to make online payment.
8. Customer should be able to rate and provide feedback.
9. Laundry shall be able to view and manage customer orders.
10. Laundry shall be able to manage services and pricing.
11. Laundry shall be able to manage delivery requests.
12. Laundry shall be able to generate reports.
13. The system shall communicate with the external delivery provider.
14. The system shall update order delivery information.

System Requirements:

1. Customer shall be able to create an account.

- 1.1 The customer shall provide full name, phone number, email, and password.
- 1.2 The system shall verify the email and phone number for uniqueness.
- 1.3 The system shall confirm successful registration through a notification.

2. Customer shall be able to log in.

- 2.1 The customer must enter a valid email and password.
- 2.2 The system shall verify the credentials before granting access.
- 2.3 The system shall redirect the customer to their home page upon successful login.

3. Customer shall be able to edit their profile information.

- 3.1 The system shall provide an option to modify personal details such as name, phone, or address.
- 3.2 The customer shall confirm any changes before saving.
- 3.3 The system shall update and store the new information.

4. Customer shall be able to specify location.

- 4.1 The system shall detect the customer's location automatically using GPS.
- 4.2 The customer can manually enter or edit their address if needed.

5. Customer shall be able to create a new laundry order.

- 5.1 The customer shall select the type of service (Washing, Ironing, Washing & Ironing).
- 5.2 The customer shall choose preferred pickup and delivery times.
- 5.3 The system shall calculate and display the total cost.
- 5.4 The customer shall confirm the order before submission.
- 5.5 The system shall send the delivery request to the external delivery provider.

6. Customer shall be able to track their order status.

- 6.1 The system shall display the current order status (Picked Up, In Progress, Ready, Delivered).
- 6.2 The system shall show delivery updates received from the external delivery provider.

7. Customer shall be able to make online payment.

- 7.1 The system shall support online payment using Mada or Apple Pay.
- 7.2 The system shall display a confirmation message once the payment is successful.

8. Customer should be able to rate and provide feedback.

- 8.1 The system should allow customers to rate their experience after order completion.
- 8.2 The system should allow customers to write comments or feedback about the laundry service

9. Laundry shall be able to view and manage customer orders.

- 9.1 The system shall display all pending, in-progress, and completed orders.
- 9.2 The laundry shall be able to accept or reject new customer orders.
- 9.3 The laundry shall be able to update each order's status (In Progress, Ready, Delivered).

10. Laundry shall be able to manage services and pricing.

- 10.1 The laundry shall be able to add, update, or remove services.
- 10.2 The laundry shall be able to modify the price of each service.
- 10.3 The system shall automatically display updated prices to customers.

11. Laundry shall be able to manage delivery requests.

11.1 The system shall send delivery requests to the external delivery provider once an order is confirmed.

11.2 The system shall update the order status when delivery updates are received from the provider.

12. Laundry shall be able to generate reports.

12.1 The system shall provide daily and monthly order reports.

12.2 The system shall display revenue summaries and performance statistics.

13. The system shall communicate with the external delivery provider.

13.1 The system shall send order details (pickup address, delivery address, and times) to the provider.

13.2 The system shall receive delivery status updates (Picked Up, Delivered) from the provider.

14. The system shall update order delivery information.

14.1 The system shall display delivery progress to the customer based on the updates received.

14.2 The system shall mark the order as “Delivered” once confirmation is received from the provider.

1.4 Non-Functional Requirements:

User Requirements:

1. The system will ensure user data security.
2. The system will support all major operating systems for smooth functionality.
3. The system will be compatible with all major web browsers.
4. The system will provide an interface that is easy to use.
5. The system will be able to handle an increasing number of users.
6. The system will be available for users at all times.

System Requirements:

1. The system will ensure user data security

- 1.1 The system must encrypt all customer, driver, and laundry data using AES-256 encryption.
- 1.2 The system must implement secure authentication using OTP or email verification.
- 1.3 The system must apply role-based access control for customers, drivers, and laundry admins.
- 1.4 The system must log all user actions for auditing and monitoring purposes.

2. The system will support all major operating systems for smooth functionality.

- 2.1 The system must run seamlessly on all major mobile and desktop operating systems.
- 2.2 The system must provide consistent performance while supporting at least 1,000 concurrent users per operating system.
- 2.3 The system must be developed using Flutter for the front-end and Node.js for the back-end to ensure cross-platform compatibility.

3. The system will be compatible with all major web browsers.

- 3.1 The system must function properly on Chrome, Firefox, Safari, and Edge, and be tested on at least 3 different versions of each browser to ensure compatibility.
- 3.2 The system must maintain layout consistency and responsive design across all browsers, and be tested on at least 5 different screen resolutions to ensure proper display.
- 3.3 The system must support at least 500 simultaneous web sessions without performance degradation or errors.

4. The system will provide an interface that is easy to use.

- 4.1 The system must have a clear and simple navigation structure.
- 4.2 The system must display all buttons, icons, and menus clearly.
- 4.3 The system must provide feedback for user actions, such as order submission or updates.
- 4.4 The system must include help tips or guidance for complex tasks.

5. The system will be able to handle an increasing number of users.

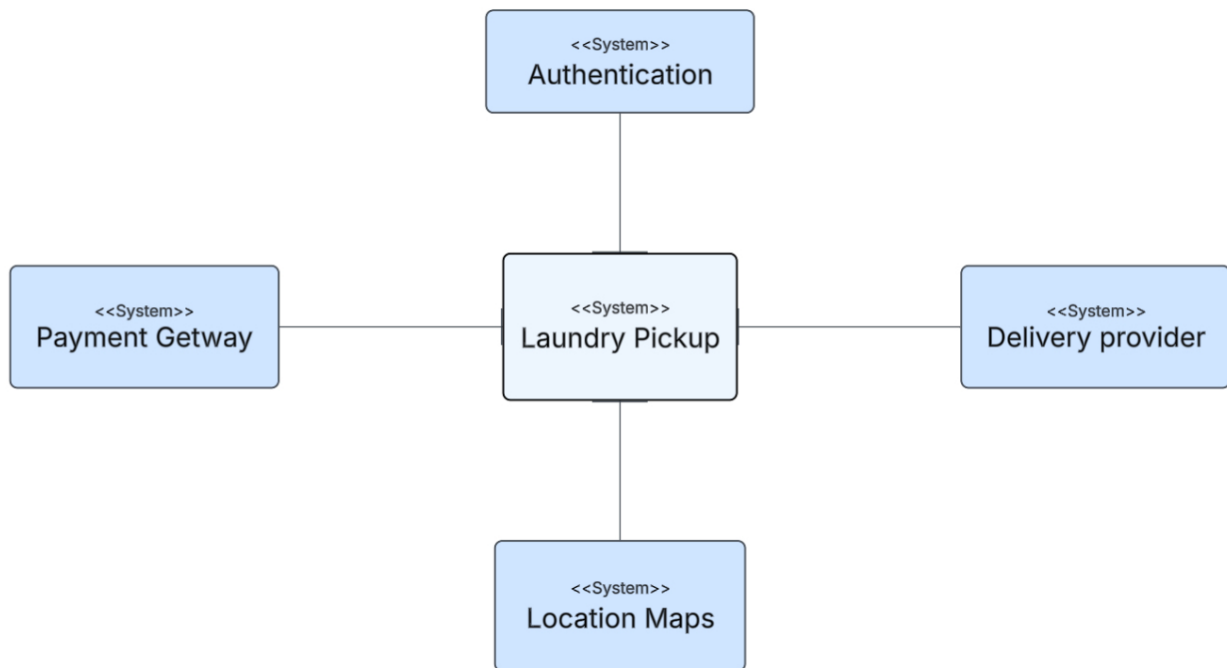
- 5.1 The system must support at least 1,500 concurrent users without performance degradation.
- 5.2 The system must efficiently manage high loads during peak times, such as weekends or holidays.
- 5.3 The system must allow for future integration of additional laundries, drivers, or services.

6. The system will be available for users at all times.

- 6.1 The system must achieve at least 99.9% uptime.
- 6.2 The system must switch automatically to backup servers in case of failure.
- 6.3 The system must perform regular database backups every 12 hours.

2. Chapter 2

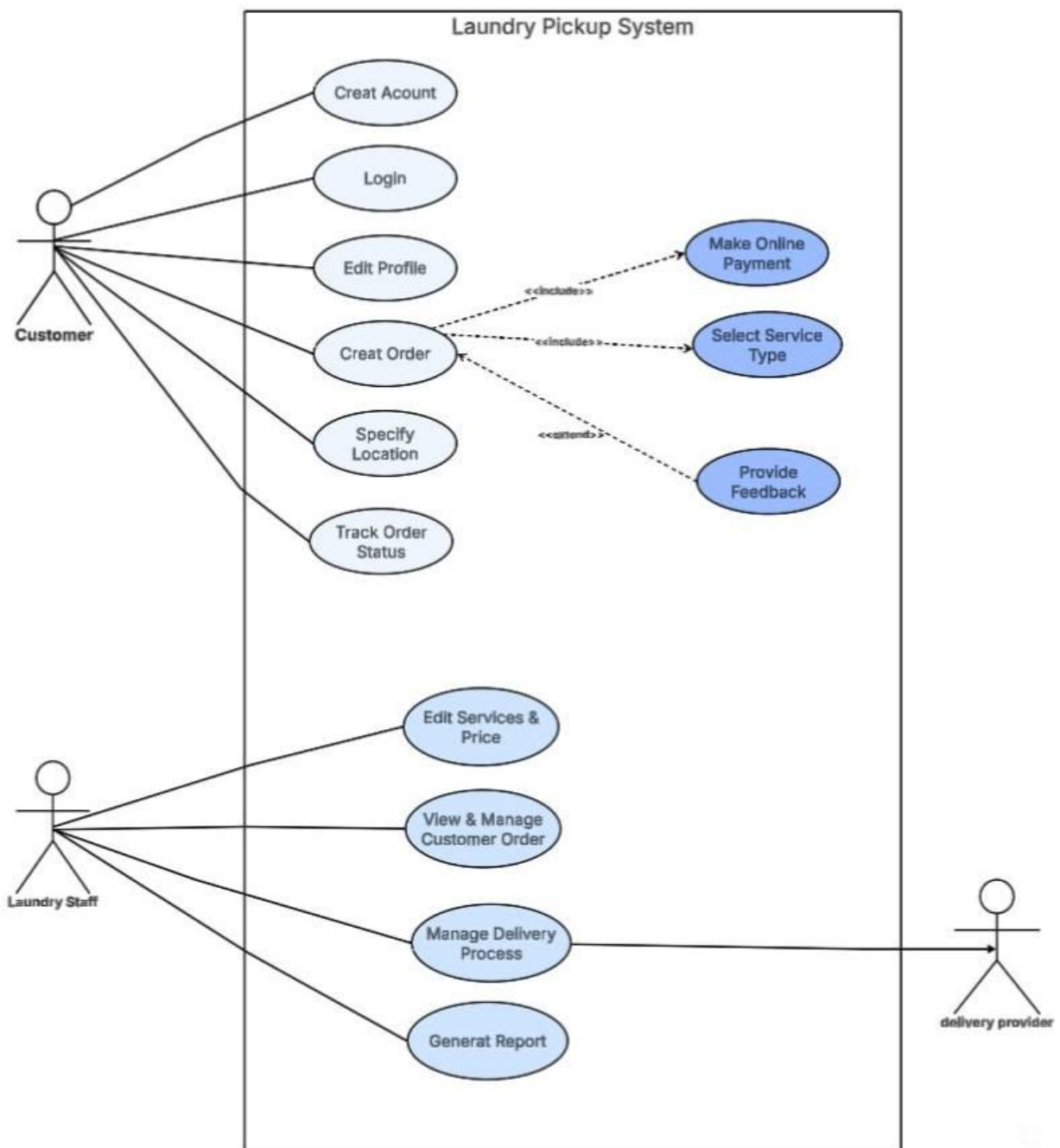
2.1 Context Diagram



2.2 System Use-case Diagram with description

In the **Laundry Pickup System**, a customer can create an account and log in. They can also edit their profile. Customers must select a service type and make an online payment before creating an order. They may optionally provide feedback after the service is completed, and they can also specify their location and track their order status.

The laundry staff can view and manage customer orders, edit services and pricing, send delivery request information to the external delivery provider, and generate reports for performance tracking. The delivery provider communicates with the system to receive and update delivery information.



2.3 Use case scenario (MSS +Extensions)

Use case 1 Create Account

Use-case Name	Create Account	
Goal	To allow the customer to register a new account in the Laundry Pickup System in order to access laundry and delivery services.	
Actors	Customer	
Main Success Scenario	1	The customer opens the Laundry Pickup System application or website.
	2	The system displays the options “Log in” and “Create Account.”
	3	The customer selects “ Create a New Account. ”
	4	The system prompts the customer to enter required details (name, email, phone number, and password).
	5	The customer provides all required information and submits the registration form.
	6	The system validates the entered information and checks for email and phone number uniqueness.
	7	If the information is valid, the system creates the account and sends a confirmation email to the customer.
	8	The customer verifies the email to activate the account.
	9	The system displays a confirmation message and redirects the customer to the login page.
Extension	6.a	If the email or phone number already exists, the system displays an error message.
	6.b	If the password does not meet security requirements, the system prompts the customer to enter a stronger password.
	8.a	If the email verification fails, the system allows the customer to request a new confirmation email.

Use case 2 Edit Profile

Use-case Name	Edit Profile	
Goal	Allow the customer to modify and update their personal details.	
Actors	Customer	
Main Success Scenario	1	The customer logs into the Laundry Pickup System.
	2	The customer navigates to the “Profile” section.
	3	The system displays the current saved details.
	4	The customer selects the information to edit (e.g., name, phone number, address).
	5	The customer confirms the details and updates the changes.
	6	The system validates the input and updates the database with the new information.
	7	The system displays a confirmation message and sends a profile update confirmation email to the customer.
Extension	6.a	If the information provided is invalid (e.g., incorrect phone format), the system prompts the customer to correct it.
	6.b	If the email address is changed, the system requires re-verification through a confirmation email before applying the update.

Use case 3 Specify Location

Use-case Name	Specify Location	
Goal	Allow the customer to set or update their pickup and delivery location.	
Actors	Customer,	
Main Success Scenario	1	The customer logs in to the system.
	2	The customer navigates to the “Location Settings” from profile Create Laundry Orderpage.
	3	The system automatically detects the customer’s location using GPS.
	4	The system displays the detected location on the map for confirmation.
	5	The customer reviews and confirms the detected location or manually edits the address if necessary.
	6	The customer saves the selected or modified location.
	7	The system validates the entered address format and stores it for future pickups and deliveries.
	8	The system displays a confirmation message indicating the successful update.
Extension	3.a	If GPS access is denied, the system prompts the customer to manually enter their address.
	7.a	If the address format is invalid or incomplete, the system requests the customer to re-enter the correct address.
	7.b	If there is a connectivity issue, the system saves the location temporarily and retries synchronization later.

Use case 4 Create Order

Use-case Name	Create Order	
Goal	To allow the customer to create a new laundry order, complete secure payment, and submit it to the Laundry Pickup System for processing.	
Actors	Customer, External Payment Gateway	
Main Success Scenario	1	The customer logs into the Laundry Pickup System.
	2	The customer selects “ Create New Order. ”
	3	The customer chooses the desired service type (Washing, Ironing, or both).
	4	The customer selects the preferred pickup and delivery times.
	5	The system calculates and displays the total cost of the order.
	6	The customer proceeds to make an online payment using a supported external payment gateway (e.g., Mada or Apple Pay).
	7	The payment gateway securely processes the transaction and sends a payment confirmation to the Laundry Pickup System.
	8	Upon receiving the confirmation, the system stores the order details (service type, location, cost, and schedule) in the database and marks the order as “ Pending. ”
	9	The system displays a confirmation message to the customer: “ <i>Your order has been successfully created.</i> ” And sends a confirmation email to the customer.
	10	After the service is completed and the order status becomes “ Delivered, ” the system allows the customer to rate and provide feedback (<i>extends → Provide Feedback</i>).
Extension	4.a	if selected time is unavailable or invalid. System asks the customer to choose another time or allow the system to suggest the nearest available time.”. Customer can accept the suggested time or select a new one manually
	6.a	If the payment gateway declines the transaction. The system displays an error message and cancels order creation.
	7.a	If no response is received from the payment gateway. The system notifies the customer that payment verification failed.
	10.a	If the customer skips the feedback. The system ends the use case normally without saving any rating.

Use case 5 Track Order Status

Use-case Name	Track Order Status	
Goal	Allow the customer to view the current status of their laundry order in real-time, including pickup, processing, and delivery updates	
Actors	Customer	
Main Success Scenario	1	The customer logs into the Laundry Pickup System.
	2	The customer select to the “My Orders” section.
	3	The system displays a list of the customer’s current and past orders.
	4	The customer selects a specific order to track.
	5	The system retrieves order details and current status from the database.
	6	The system displays the order status, including: • Picked Up • In Progress • Ready for Delivery • Out for Delivery • Delivered
	7	The customer can refresh the status at any time to see real-time updates.
Extension	4.a	If the order ID or reference is invalid, the system displays: “Order not found. Please check your order number or contact support.”
	5.a	If the Order data not available system cannot retrieve order status due to a technical issue or connectivity problem, it displays: “Unable to retrieve order status at this time. Please try again later.”

Use case 6 Manage Services and Pricing

Use-case Name	Manage Services and Pricing	
Goal	Allow laundry staff to add, update, or remove available services and modify their prices.	
Actors	Laundry Staff	
Main Success Scenario	1	The laundry staff logs into the system.
	2	The staff select to the “Services Management” section.
	3	The system displays a list of all available services and their prices.
	4	The staff can add a new service by entering the service name, description, and price.
	5	The staff can edit existing services or update prices.
	6	The system validates the new or updated information.
	7	The system updates the database and displays a confirmation message.
	8	The updated list becomes immediately visible to customers.
Extension	4.a	If the service name already exists, the system displays an error message.
	6.a	If invalid input is provided (e.g., negative price), the system prompts the staff to correct it.
	8.a	If database synchronization fails, the system temporarily stores the changes and alerts the staff.

Use case 7 View & Manage Customer Order

Use-case Name	View & Manage Customer Order	
Goal	Allow laundry staff to view all customer orders and manage their processing stages.	
Actors	Laundry Staff	
Main Success Scenario	1	The laundry staff logs into the system.
	2	The system displays the dashboard with all current orders.
	3	The staff views order details including order ID, customer name, and service type.
	4	The staff reviews pending orders and accepts or rejects them.
	5	When an order is accepted, the system updates its status to “In Progress.”
	6	Once the service is completed, the staff updates the order status to “Ready.”
	7	The system displays a confirmation message for each update.
	8	After confirming delivery, the staff marks the order as “Delivered.”
	9	The system sends an email notification to the customer about the updated order status.
Extension	4.a	If the staff rejects an order, the system changes its status to “Cancelled” and notifies the customer.
	9.a	If the email notification fails, the system logs the update internally and displays an alert to retry later.

Use case 8 Manage Delivery Process

Use-case Name	Manage Delivery Process	
Goal	Allow laundry staff to manage both pickup and delivery requests by coordinating with the external delivery provider using accurate order and location details.	
Actors	Laundry Staff, External Delivery Provider	
Main Success Scenario	1	The laundry staff logs into the system.
	2	The staff select to the “Delivery Management” page.
	3	The system displays two sections: Pending Pickups and Pending Deliveries.
	4	For pending pickups, the system shows requests automatically generated after customer order confirmation, including pickup address, delivery address, pickup time, and delivery time.
	5	The staff verifies that the pickup and delivery details are correct and that the order is ready to be processed.
	6	The staff confirms the pickup request through the system.
	7	The system sends the pickup details (addresses and time) to the external delivery provider.
	8	Once the provider confirms the pickup scheduling, the system updates the order status to “Scheduled for Pickup.”
	9	After receiving a confirmation from the provider that the laundry has been collected, the system automatically updates the status to “Picked Up.”
	10	Once the laundry process is complete, the staff marks the order as “Ready for Delivery.”
	11	The staff submits a delivery request through the system, which includes delivery address and time for the return.
	12	The delivery provider confirms scheduling, and the system updates the order status to “Scheduled for Delivery.”
	13	After the provider completes the delivery, the staff updates the order status to “Delivered.”
	14	The system records the completion and sends a confirmation email to the customer.
Extension	7.a	If communication with the delivery provider fails, the system queues the request and notifies the staff to retry later.
	8.a	If pickup scheduling is delayed by the provider, the system marks the order as “Awaiting Pickup Confirmation.”
	11.a	If the selected delivery time is unavailable, the system suggests the nearest available time.
	13.a	If delivery confirmation is not received within a set period, the system flags the order as “Pending Delivery Confirmation.”
	14.a	If the confirmation email to the customer fails to send, the system logs the issue and shows an in-app confirmation message instead.

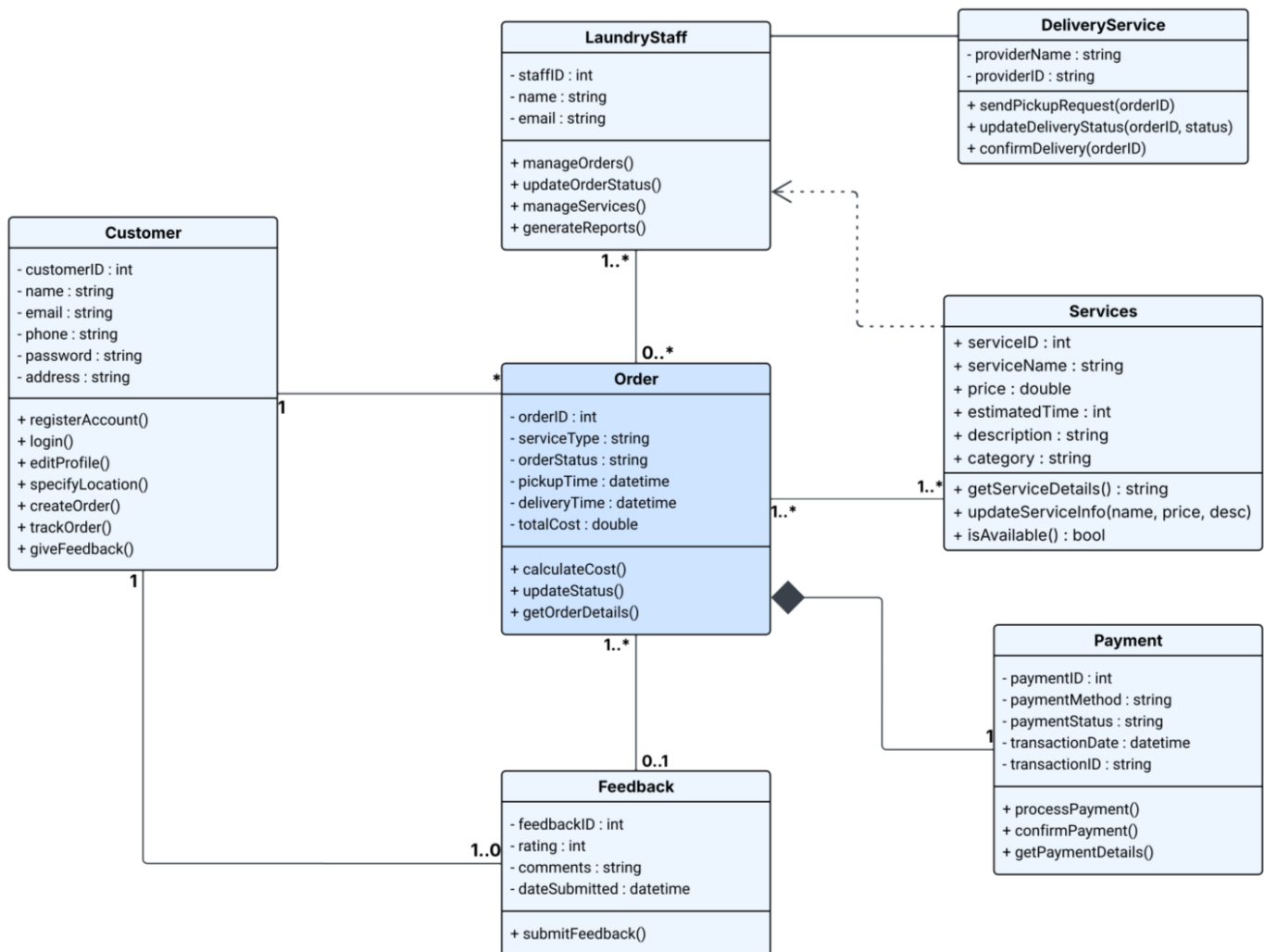
Use case 9 Generate Reports

Use-case Name	Generate Reports	
Goal	Allow laundry staff to generate daily or monthly reports summarizing order volume, revenue, and performance.	
Actors	Laundry Staff	
Main Success Scenario	1	The laundry staff logs into the system.
	2	The staff select to the “Reports” section.
	3	The system displays available report types (Daily, Monthly).
	4	The staff selects the desired report period and parameters (e.g., date range, service type).
	5	The system retrieves data from completed orders.
	6	The system generates a report showing total orders, revenue, and performance statistics.
	7	The staff can export the report in PDF or Excel format.
	8	The system displays a success message confirming the report generation.
Extension	4.a	If invalid parameters are entered, the system prompts the staff to reselect valid values.
	5.a	If data retrieval fails, the system notifies the staff and allows retrying.
	7.a	If exports fail, the system allows downloading the report later from the archive section.

3. Chapter 3

3.1 System Class Diagram

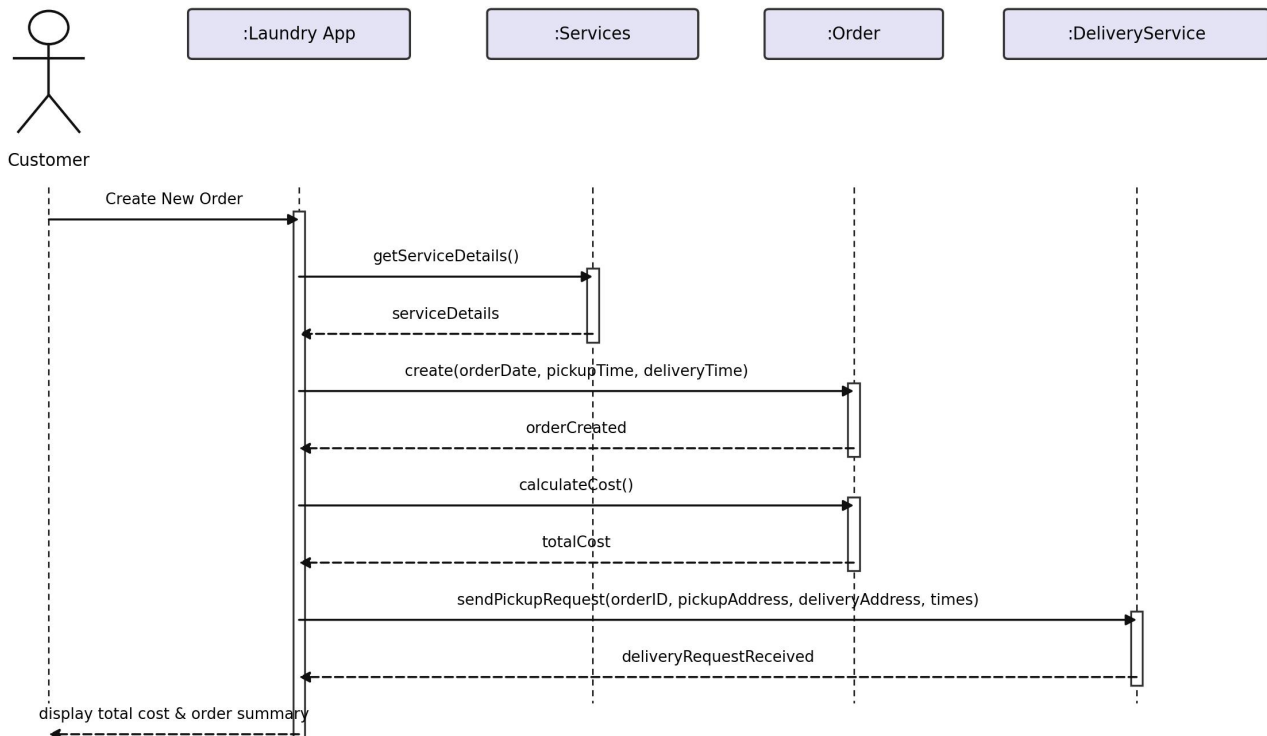
The class diagram represents the structure of the Laundry Pickup System and shows how its main components interact. A Customer can create multiple Orders, each containing selected Services, a Payment record, and an associated delivery process through DeliveryService. LaundryStaff manages orders and services within the system. The Feedback class allows customers to submit ratings for completed orders. Overall, the diagram highlights how orders act as the central link connecting customers, payments, services, delivery operations, and feedback.



3.2 Sequence Diagrams

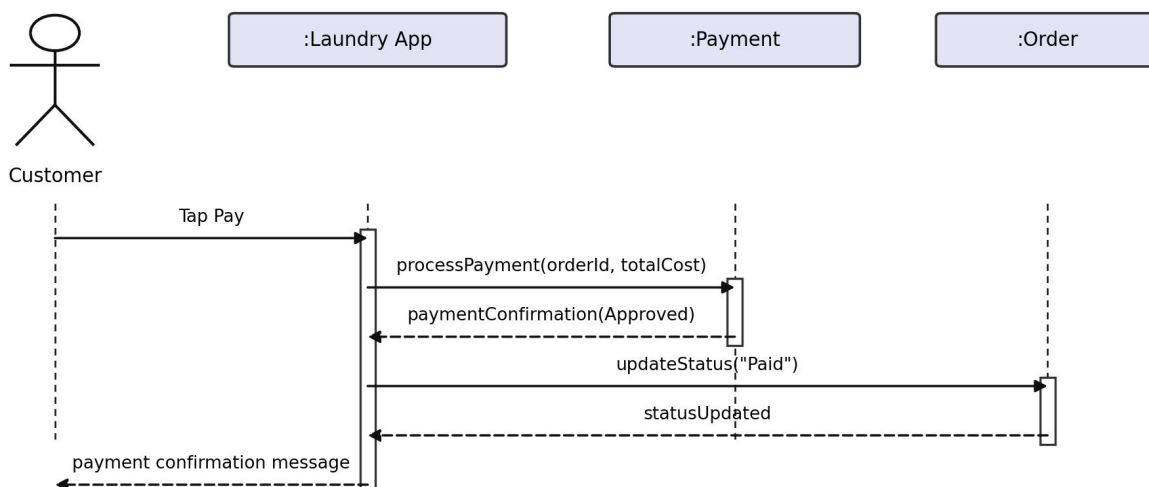
Sequence diagram for Create New Order

Sequence Diagram for Create New Order



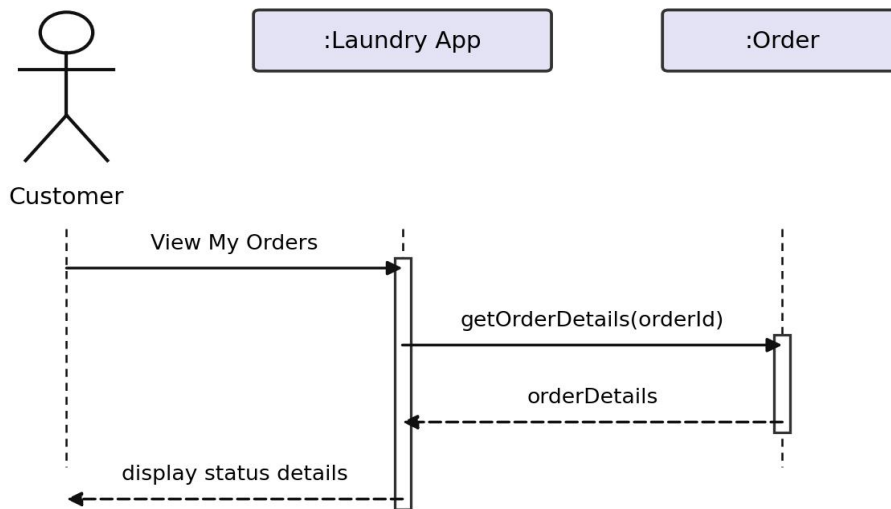
Sequence diagram for Pay via External Payment System

Sequence Diagram for Pay via External Payment System



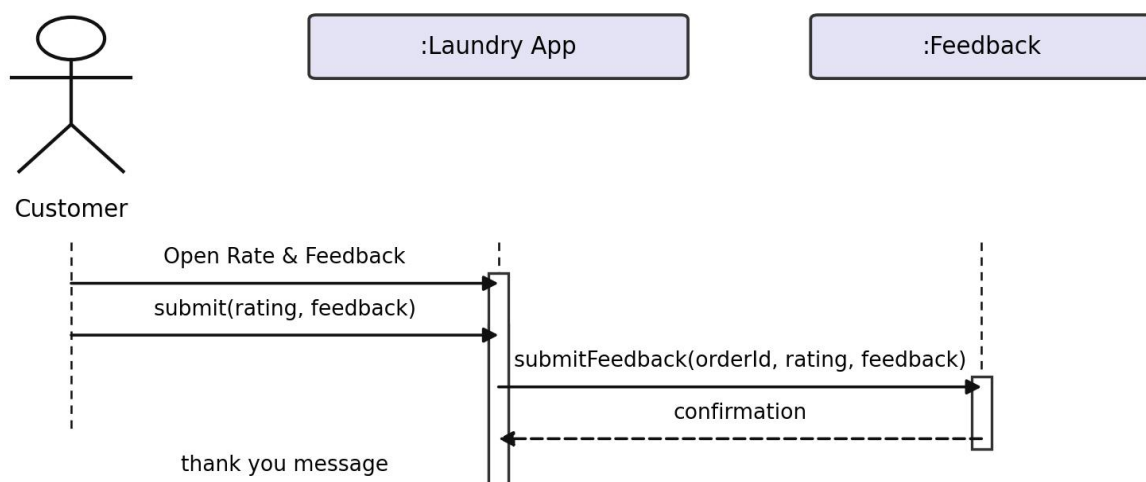
Sequence diagram for Track Order Status

Sequence Diagram for Track Order Status

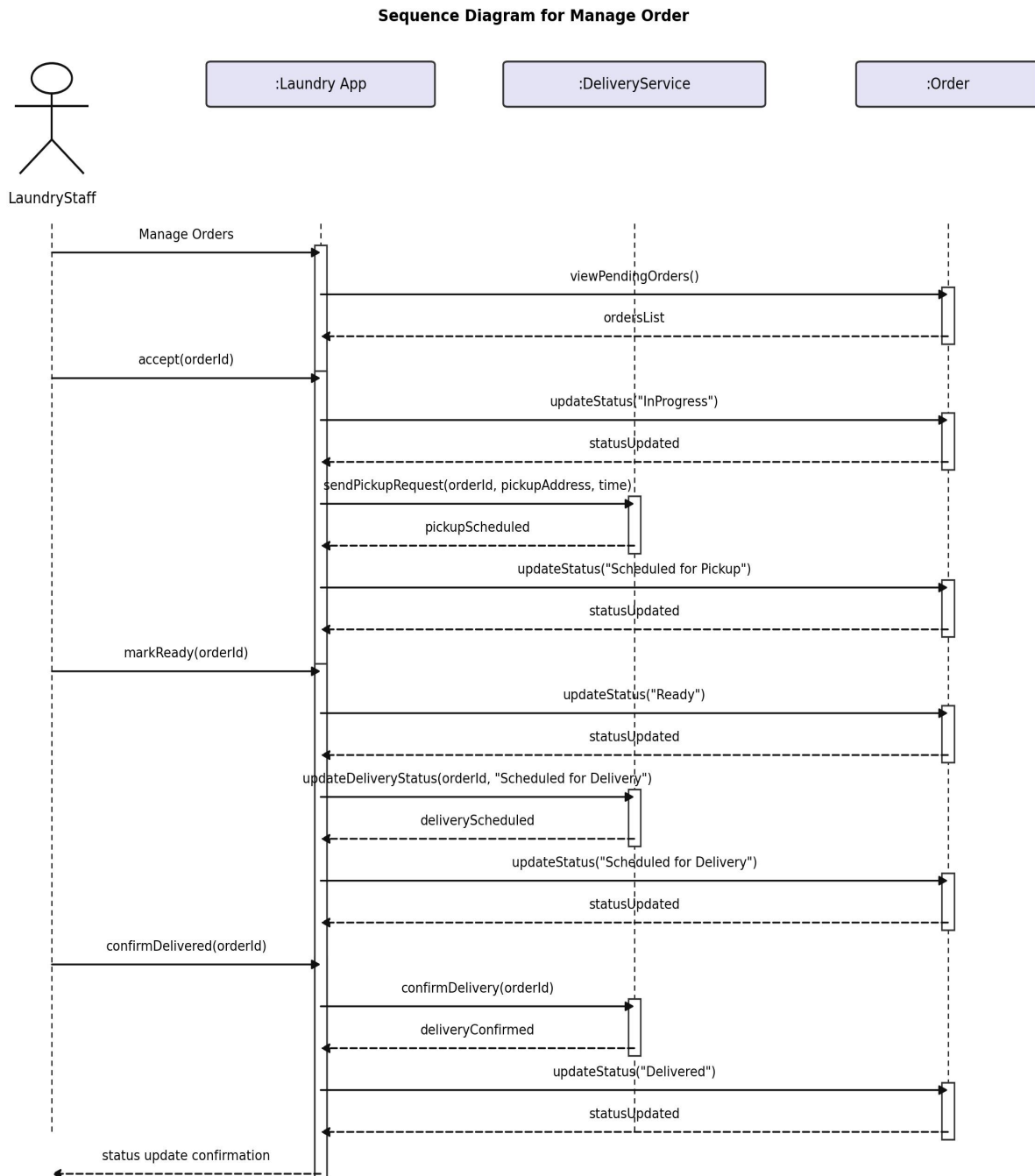


Sequence diagram for Rate & Feedback

Sequence Diagram for Rate & Feedback



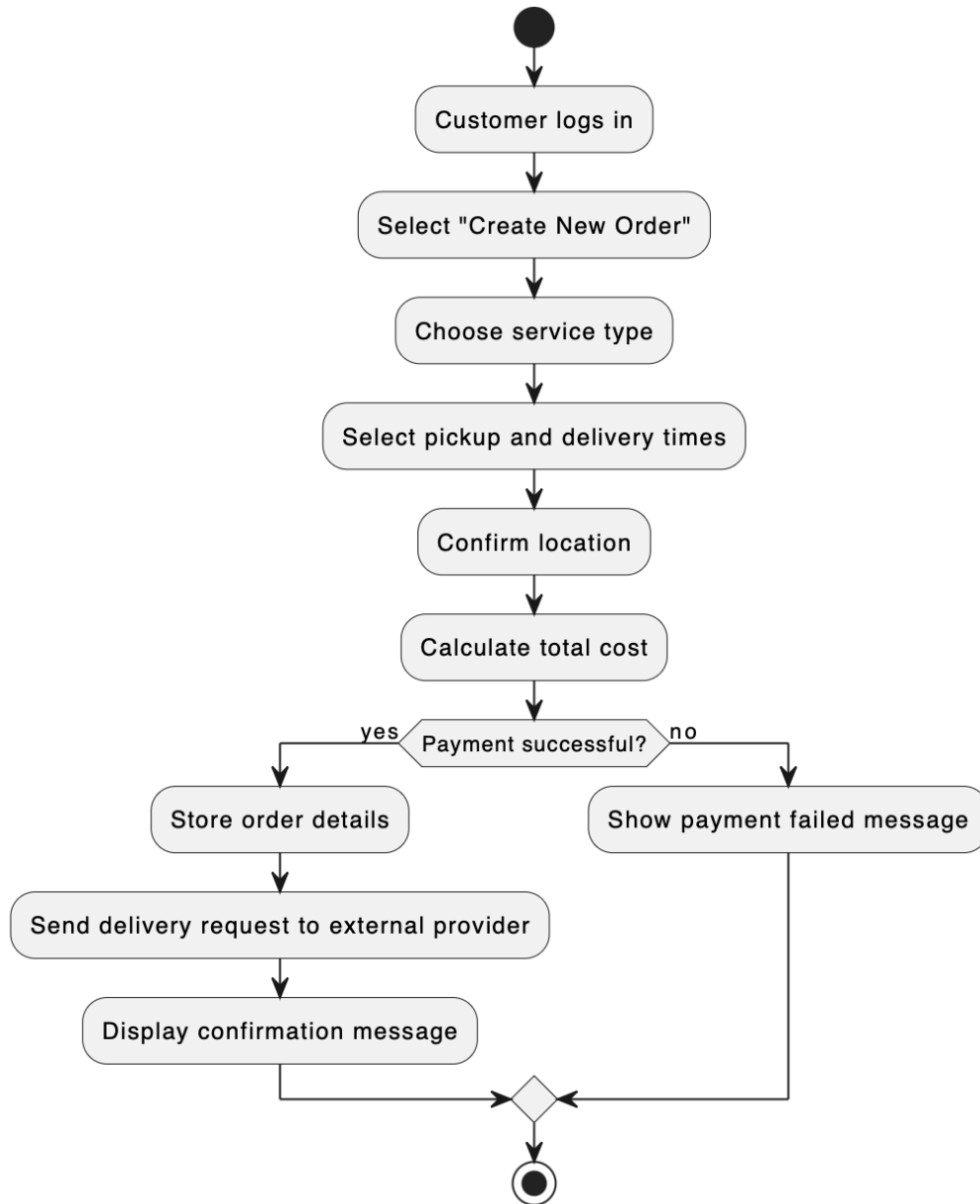
Sequence diagram for Manage order



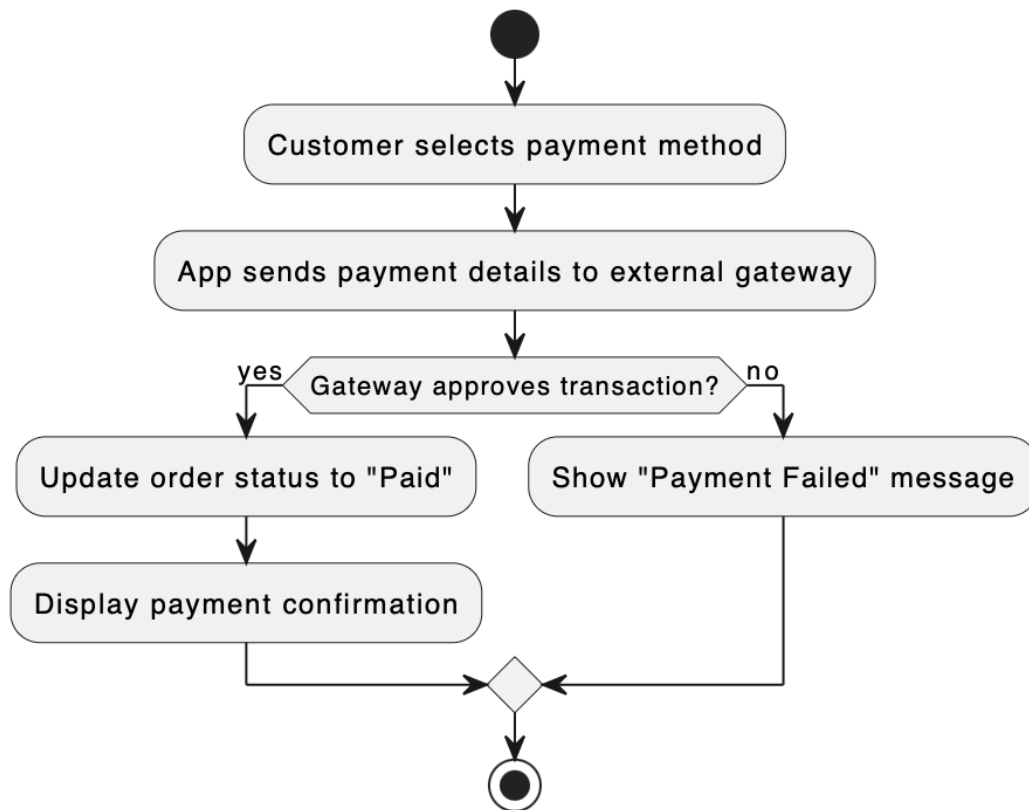
4. Chapter 4

4.1 Activity Diagram

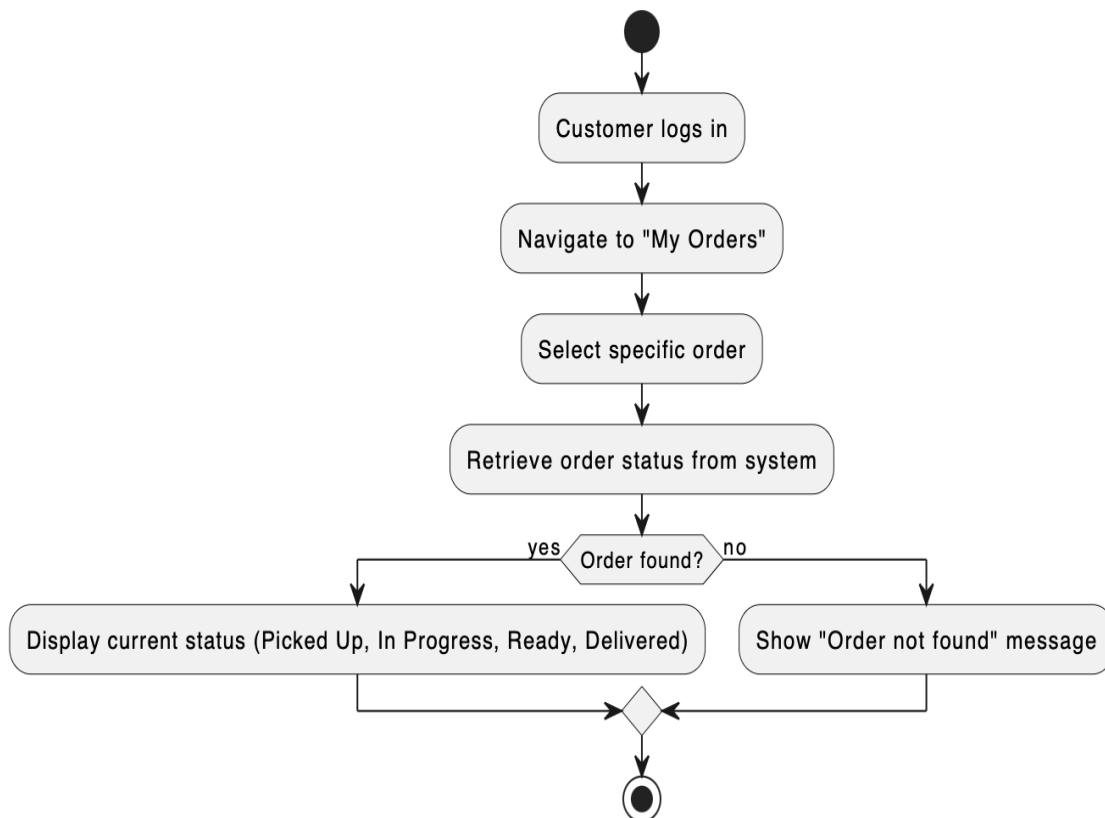
Activity Diagram for Create New Order



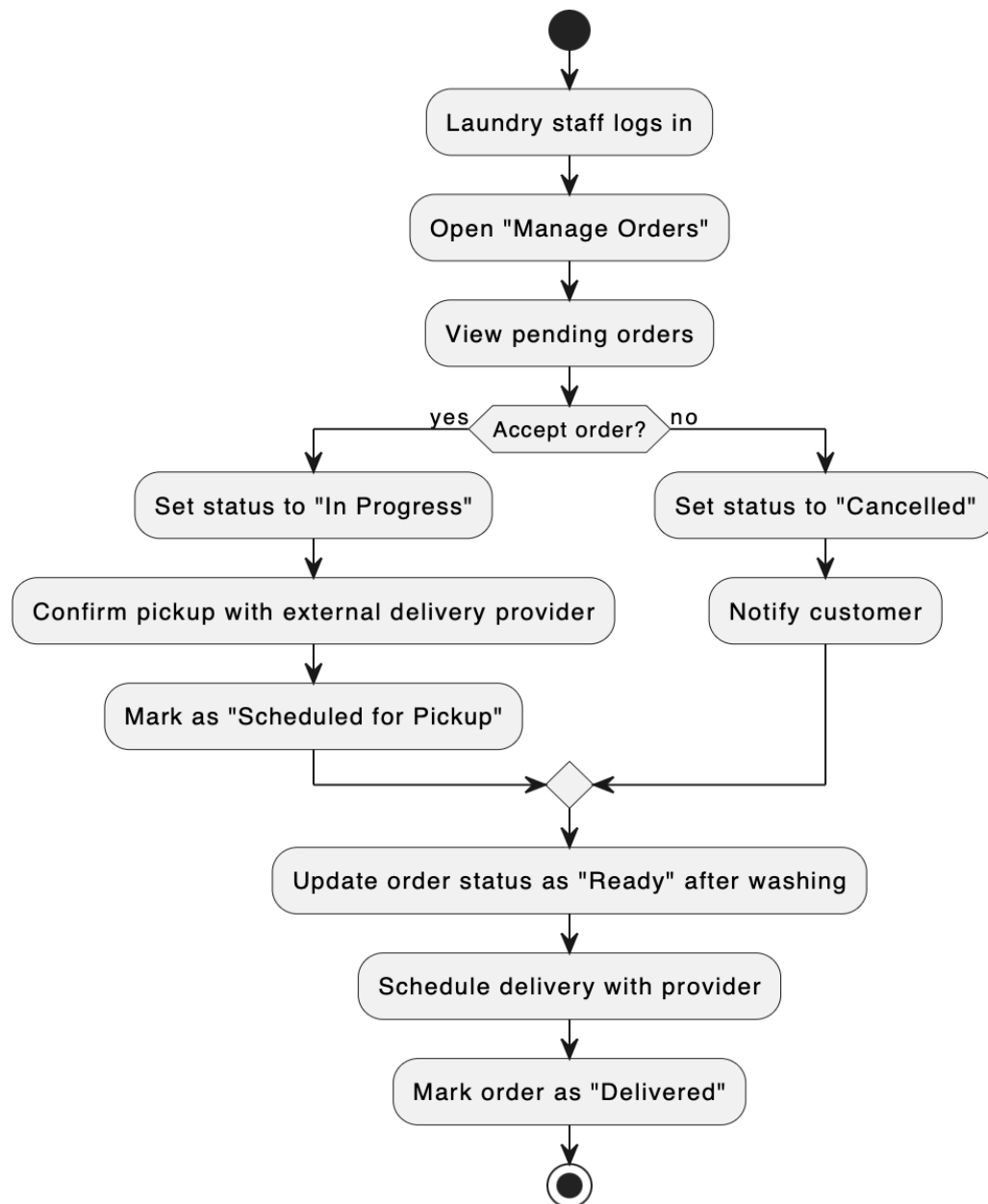
Activity Diagram for Pay via External Payment System



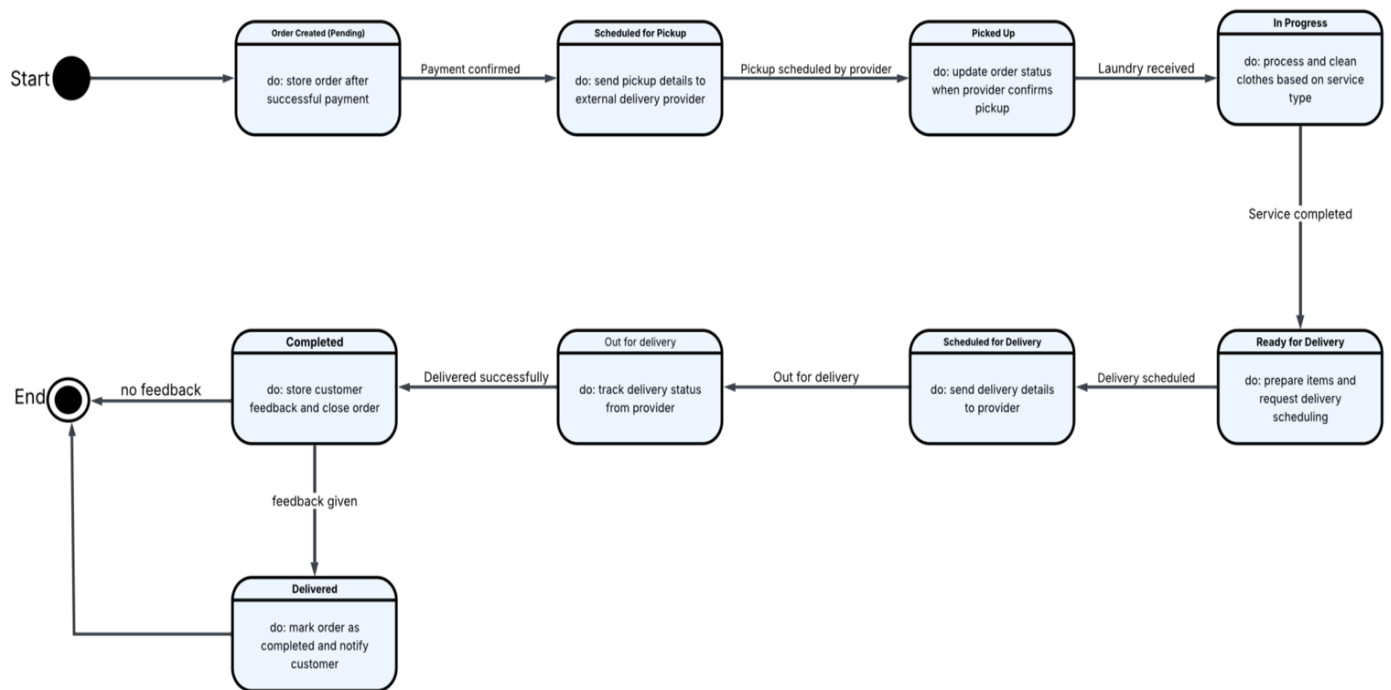
Activity Diagram for Track Order Status



Activity Diagram for Manage Order (Laundry Staff)



State Diagram for Order status



5. Chapter 5

5.1 Architecture Diagram

We selected the **Client-Server Architecture** for our project because it clearly separates the client interface from the server logic, improving **performance** and organization. This structure enhances **security** by keeping data and processing on the server side and **ensures easy maintenance** through centralized updates. It also supports **scalability** as more users or features can be added with minimal effort, making it a suitable and efficient choice for the Laundry Pickup System.

